

5. Mean speed is defined by the equation:

$$\text{mean speed} = \frac{\text{total distance}}{\text{total time}}$$

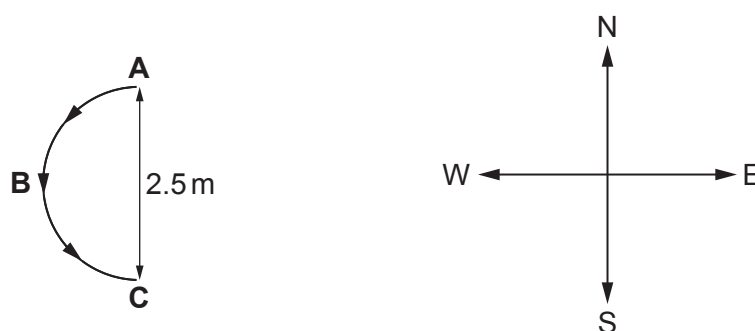
(a) State the definition of mean **velocity**.

[1]

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(b) Ieuan and Seren play with their toy train. A train takes 4.0 s to travel at constant speed in a semicircle **ABC** as shown.



(i) Show that the train's speed is approximately 1 ms^{-1} .

[1]

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(ii) Calculate the mean velocity of the train as it moves from **A** to **C**.

[2]

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(iii) Ieuan believes that, since the speed between **A** and **C** is constant, the train is not accelerating. Seren disagrees. Explain who is correct, justifying your answer. [3]

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- (c) **From C**, the train continues in a straight line with a speed of 1.0 m s^{-1} for 2.0 s . It then decelerates uniformly to rest in a distance of 1.2 m . Complete the velocity-time graph and the displacement-time graph **for the motion from C** on the grids below. Space for calculations. [6]

